

MSSC2025
TORINO 8-12 SEP 2025



Local Gaussian-type basis functions for solids



**UNIVERSITÀ
DI TORINO**

Bartolomeo Civalleri

Department of Chemistry, Via P. Giuria 7

University of Torino

bartolomeo.civalleri@unito.it



Outline



- What basis functions for solids?
- From Atomic Orbitals to Gaussian-type functions
- Gaussian basis sets (form, classification, extensions, notation, ...)
- How to choose a Gaussian basis set
- Databases and libraries
- All-electron or *effective core potentials* (*ECP*)

One electron equations

$$H\psi_i = \left(-\frac{1}{2}\nabla^2 + V_{en} + V_H + V_{eff}\right)\psi_i = \varepsilon_i\psi_i$$

$$V_{eff}^{DFT}\psi_i = F\left[n(\vec{r}), \nabla n(\vec{r}), \tau(\vec{r}), \nabla^2 n(\vec{r}), \dots\right]\psi_i$$

$$\begin{aligned} V_{eff}^{HF}\psi_i &= \int dr_2 V(\vec{r}_1, \vec{r}_2)\psi_i(\vec{r}_2) \\ &= -\sum_j \int dr_2 \frac{\psi_j^*(\vec{r}_2)\psi_i(\vec{r}_2)\psi_j(\vec{r}_1)}{|\vec{r}_1 - \vec{r}_2|} \end{aligned}$$

$\{\psi(\mathbf{r}_i)\}=?$

A representation for $\psi(r_i)$

Alternatives

1. **Direct representation on a grid** - $\psi(r_i)$ (Numerical approach)
Potentially very accurate but huge grids required – expensive
2. **Expansion in a basis set** (Algebraic approach)
*Compact representation, potential for high accuracy, efficient,
choose functions for computational convenience.
Solve by iteration and diagonalisation (SCF) or direct minimisation.*

$$\psi_i(\mathbf{r}) = \sum_{\mu}^{N_{BS}} c_{\mu i} \chi_{\mu}(\mathbf{r})$$

$$H_i \psi_i = \left(-\frac{1}{2} \nabla_i^2 + V_{en} + V_H + V_{eff} \right) \psi_i = \varepsilon_i \psi_i \quad \longrightarrow \quad \text{Matrix equation} \quad \mathbf{HC} = \mathbf{SCE}$$

$$H_{\mu\nu} = \langle \chi_{\mu} | H_i | \chi_{\nu} \rangle \quad \text{Hamiltonian matrix} \quad S_{\mu\nu} = \langle \chi_{\mu} | \chi_{\nu} \rangle \quad \text{Overlap matrix}$$

$$H_{\mu\nu} = \left\langle \chi_{\mu} \left| -\frac{1}{2} \nabla_i^2 + V_{en} \right| \chi_{\nu} \right\rangle + \sum_{\lambda\rho}^{N_{BS}} D_{\lambda\rho} \langle \chi_{\mu} \chi_{\lambda} | V_H | \chi_{\nu} \chi_{\rho} \rangle + \langle \chi_{\mu} | V_{eff} | \chi_{\nu} \rangle$$

Expansion in a basis set

One-electron functions are expressed as a linear-combination (LC) of known *basis functions* $\{\chi_i\}$ denoted as **basis set**:

$$\psi_i(\mathbf{r}) = \sum_{\mu}^{N_{BS}} c_{\mu i} \chi_{\mu}(\mathbf{r})$$

Ideally, basis functions should:

1. allow *orderly and systematic extension* towards completeness with respect to one-electron functions (complete basis set limit: $N_{BS} \rightarrow \infty$)
2. allow *rapid convergence* for a reasonably accurate description of solid electronic bands and charge distributions
3. have an *analytical form* for easy and accurate calculation of one- and two-electron integrals
4. *be orthogonal* or at least their non-orthogonality should not pose problems related to numerical instability

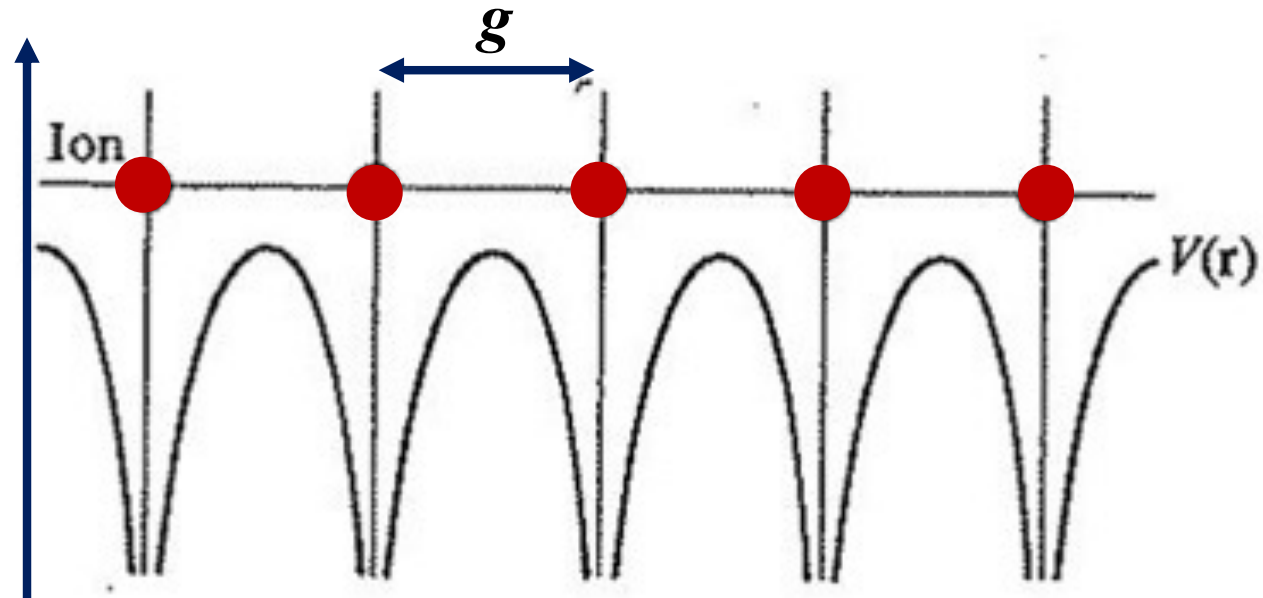
Electrons in a periodic potential

Periodic potential

$$V(\mathbf{r}) = V(\mathbf{r} + \mathbf{g})$$

Bloch's theorem

$$\psi_i^{\mathbf{k}}(\mathbf{r}) = u_{\mathbf{k}}(\mathbf{r}) e^{i\mathbf{k} \cdot \mathbf{r}}$$



Solid state physics: nearly free-electron model (weak periodic potential)

Electrons are free to move as “waves” -> *free electron-like plane wave levels*

Solid state chemistry: tight-binding model (highly localized atomic states)

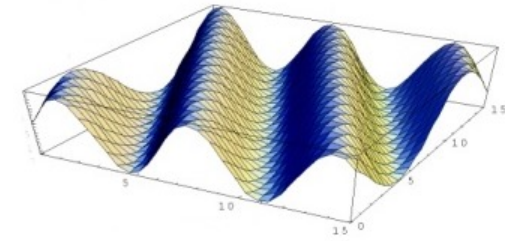
Electrons are more “localized” around the nucleus -> *atomic-like features*

Bloch functions: **Plane-waves**

Each crystalline orbital is a linear combination of Bloch functions which are defined in terms of

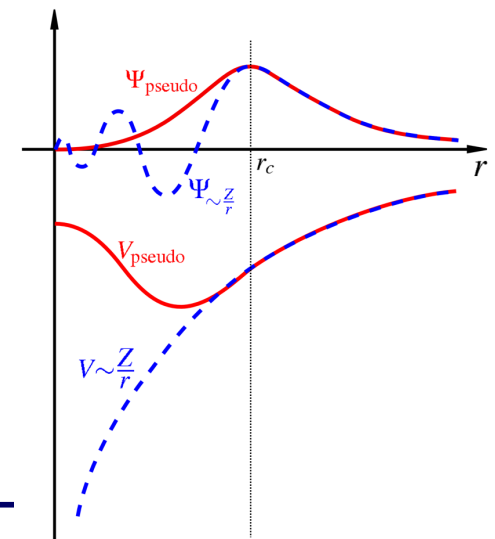
Plane-waves (PW):

$$\psi_i(\mathbf{r}; \mathbf{k}) = \sum_{\mathbf{K}}^{(\mathbf{k}+\mathbf{K})^2 < E_{cut}} c_{\mathbf{K},i}(\mathbf{k}) e^{-i(\mathbf{k}+\mathbf{K})\mathbf{r}}$$



(e.g. QE, VASP, CASTEP)

- Plane waves are inherently Bloch functions (a 3D box is needed)
- Complete set of **orthogonal** basis functions.
- The number of plane waves and the basis-set quality is controlled by a single energy-cutoff value (E_{cut}).
- Systematic improvement with E_{cut}
- It is very difficult to describe the electronic structure near the nuclei.
- **Pseudopotentials** must be used.



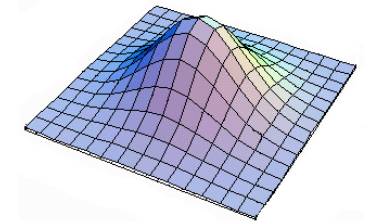
Bloch functions: Localized functions

Each crystalline orbital is a linear combination of Bloch functions which are defined in terms of

Localized functions

$$\psi_i(\mathbf{r};\mathbf{k}) = \sum_{\mu} c_{\mu,i}(\mathbf{k}) \phi_{\mu}(\mathbf{r};\mathbf{k})$$

$$\phi_{\mu}(\mathbf{r};\mathbf{k}) = \sum_{\mathbf{g}} \chi_{\mu}(\mathbf{r} - \mathbf{R}_{\mu} - \mathbf{T}) e^{i\mathbf{k} \cdot \mathbf{T}}$$

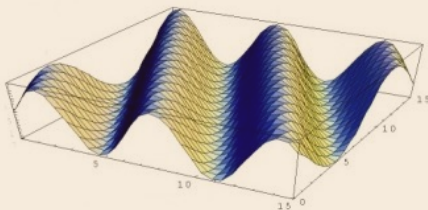


(e.g. **CRYSTAL**)

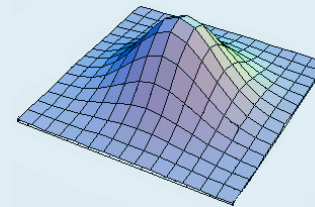
where $\mathbf{R}_{\mu}$ is the position of the function in the unit cell $\mathbf{T}$.

- Basis functions are usually localized at the nuclear positions
- They mimic *atomic-type orbitals* (AO) by means of **Gaussian-type functions**
- This corresponds to the **LCAO approximation (CO-LCAO)**
- Complete set of **nonorthogonal** basis functions
- **All-electron** calculations or *effective core potentials (ECP)* for heavy atoms
- Basis set convergence can be slow (*Basis Set Incompleteness Error, BSIE*)
- Easier implementation of HF and hybrid functionals (*slightly more costly than GGA*)

Plane Waves vs Gaussian BS



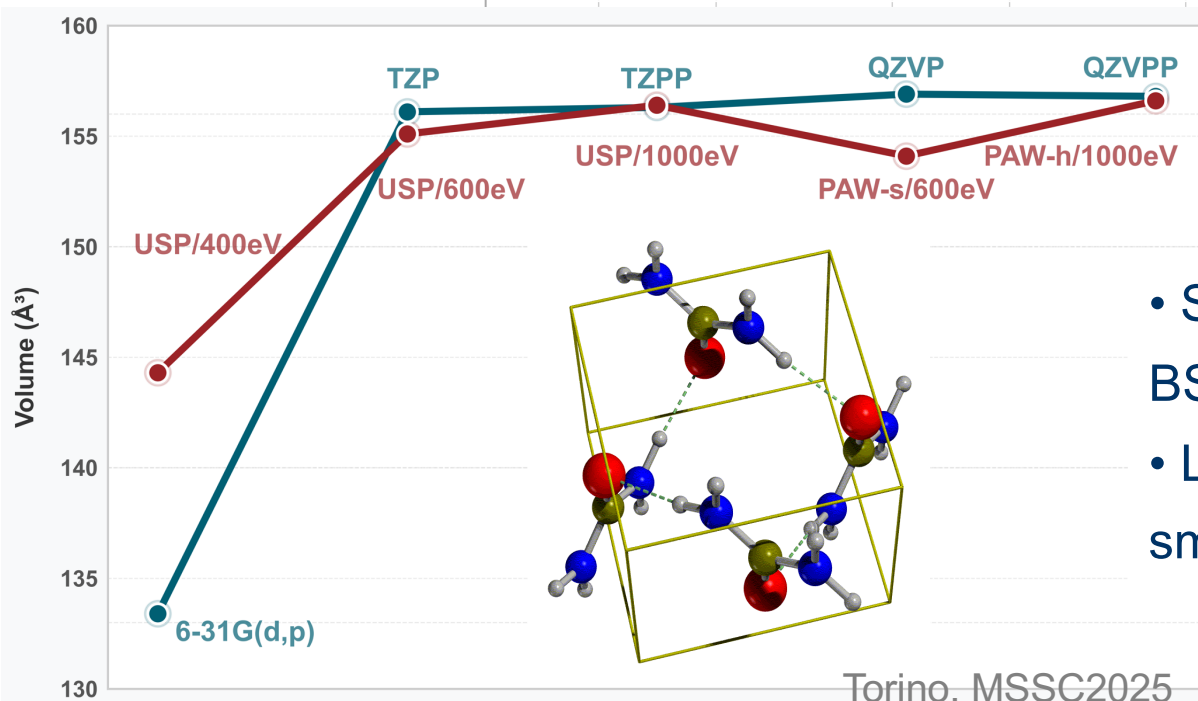
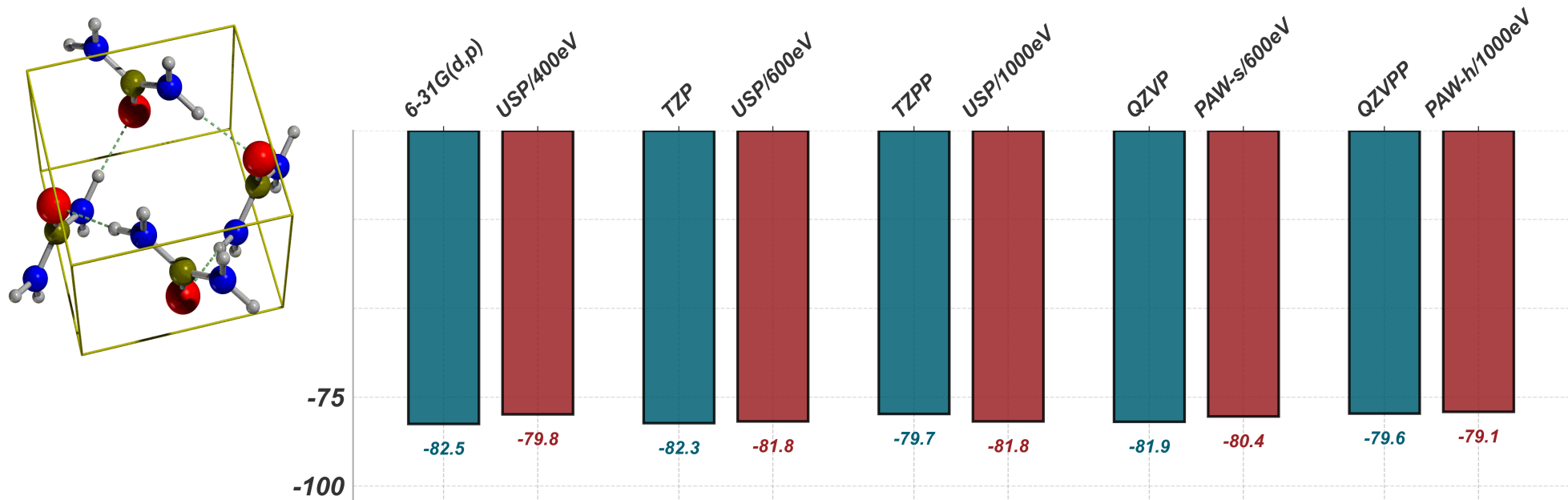
- + Universal
- Use of pseudopotentials is needed
- Large number of PW needed
- Non local
- + One parameter sufficient (E_{cut})
- + Computational effort increases slowly
- Loss of chemical insight
- = Pulay stress
- = Numerical efficient (FFT)
- 3D periodic replica for 2D, 1D, 0D sys.s
- + No BS Superposition Error (BSSE)
- Hybrid-DFT calculations are costly



- Element specific
- + All-electron and ECP
- + Small GBS can give accurate results
- + Local
- Systematic improvement not easy
- Computational effort increases quickly
- + Chemical insight is given
- = Pulay forces
- = Computational efficient (integrals)
- + No artificial replica for 2/1/0D sys.s
- BS Errors: BSIE and BSSE
- + Hybrid-DFT calculations are slightly more costly than (m)GGA

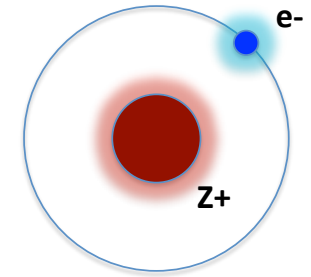
Comparing GTFs and PWs: Urea crystal - PBE

Cohesive Energy (kJ/mol)



- Small GTF basis sets (large BSSE) give too compact structures
- Lower Ecut in PWs calc.s leads to smaller unit cells (Pulay stress)

Hydrogenic Orbitals (HO)



Solution of the Schödinger equation for hydrogen-like atoms

General form:

$$\chi_{nlm}(r, \theta, \varphi) = R_{nl}(r)Y_{lm}(\theta, \varphi)$$

Radial function

$$R_{nl}(r) = N_{nl}\rho^l L_{n+1}^{2l+1}(\rho) e^{-\rho/2}$$

$L_{n+1}^{2l+1}(\rho)$ Laguerre polynomials

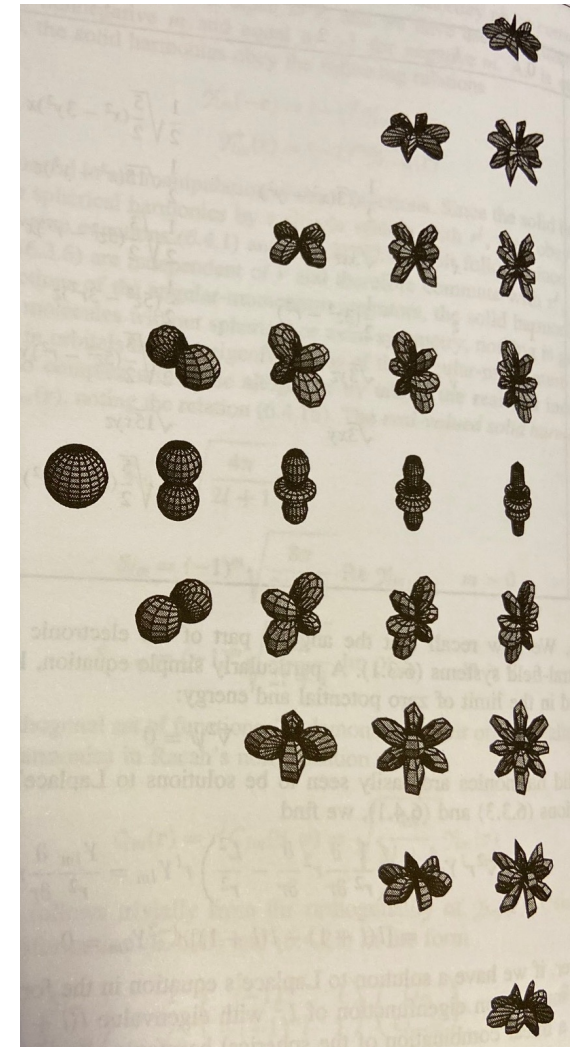
$$\rho = \frac{2Z}{n}r$$

Angular function
(Spherical Harmonics)

$$Y_{lm}(\theta, \varphi)$$

Real Spherical Harmonics
e.g. $0 < l < 4$; $m=0, \dots, \pm l$

- AO are at the basis of the Molecular Orbital theory
- Atom centered basis functions
- Unfortunately, **NOT** ideal for LCAO calculations



Localized basis functions

Localized functions retain the product form but adopt different radial functions

$$\chi_{nlm}(r, \theta, \varphi) = R_{nl}(r) Y_{lm}(\theta, \varphi)$$

Slater-Type Orbitals (STO)

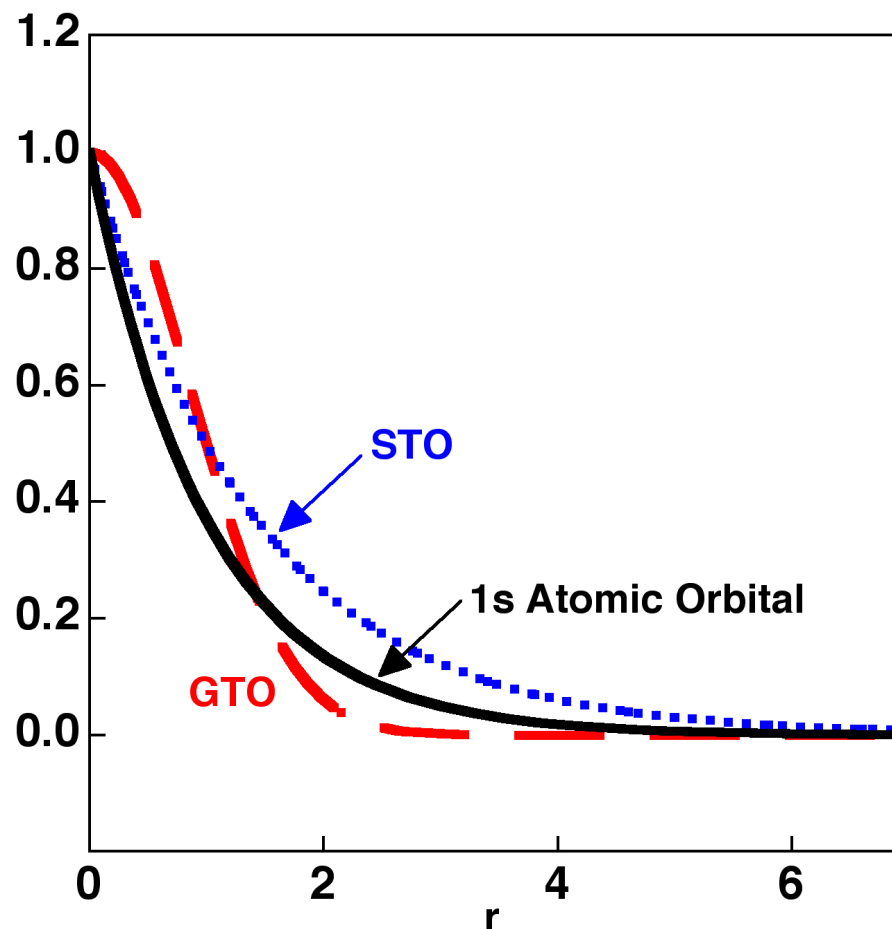
$$R_{nl}^{STO}(r) = N_{nl} r^{n-1} e^{-\zeta r}$$

HO-like behaviour. Fast convergence.
Core region properly described (Cusp at nucleus). Not ideal for integrals

Gaussian-Type Functions (GTF)

$$R_{nl}^{GTF}(r) = N_{nl} r^{2n-l-2} e^{-\alpha r^2}$$

Wrong behaviour (Max. at nucleus – Too fast decay). Slow convergence.
Easier to calculate 2e-integrals



Gaussian-type basis functions

- **Spherical Gaussian functions**

$$\chi_{nlm}^{\text{Spher}}(r, \theta, \varphi) = N_{nlm} Y_{lm}(\theta, \varphi) r^{2n-l-2} e^{-\alpha r^2}$$

- **Cartesian Gaussian functions**

$$\chi_{nlm}^{\text{Cart}}(r, \theta, \varphi) = N_{nlm} x^{l_x} y^{l_y} z^{l_z} e^{-\alpha r^2}$$

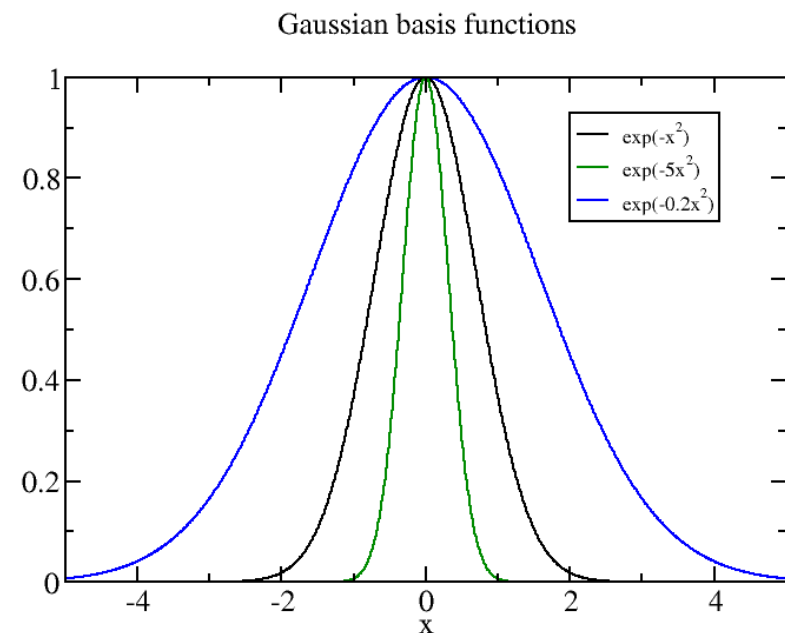
For cartesian functions, l_x , l_y and l_z determine the type of orbitals (they are not quantum numbers)

Redundancy: 6d vs 5d, 10f vs 7f, ... func.s

Computational advantages

The product of two Gaussian functions is another Gaussian function.

GTF are characterized by the value of α (Bohr⁻²)



Large α : tight - **Small α : diffuse**

Analytic and efficient calculation of electronic repulsion integrals
Efficient pre-screening (locality)

Primitive and Contracted Gaussian functions

To recover the correct “cusp” behaviour at the nucleus and describe the core region well, GTF are usually constructed as a fixed linear combination (*contraction*) of **Primitive Gaussian functions** (G)

$$\chi_{\mu}(\mathbf{r} - \mathbf{R}_{\mu} - \mathbf{T}) = \sum_j^{N_G} d_{\mu,j} G(\mathbf{r} - \mathbf{R}_{\mu} - \mathbf{T}; \alpha_{\mu,j})$$

$$G(\mathbf{r} - \mathbf{R}_{\mu} - \mathbf{T}; \alpha) = N_{nlm} x^{l_x} y^{l_y} z^{l_z} e^{-\alpha r^2}$$

$\alpha_{\mu,j}$: contraction **exponents**
 $d_{\mu,j}$: **contraction coefficients**

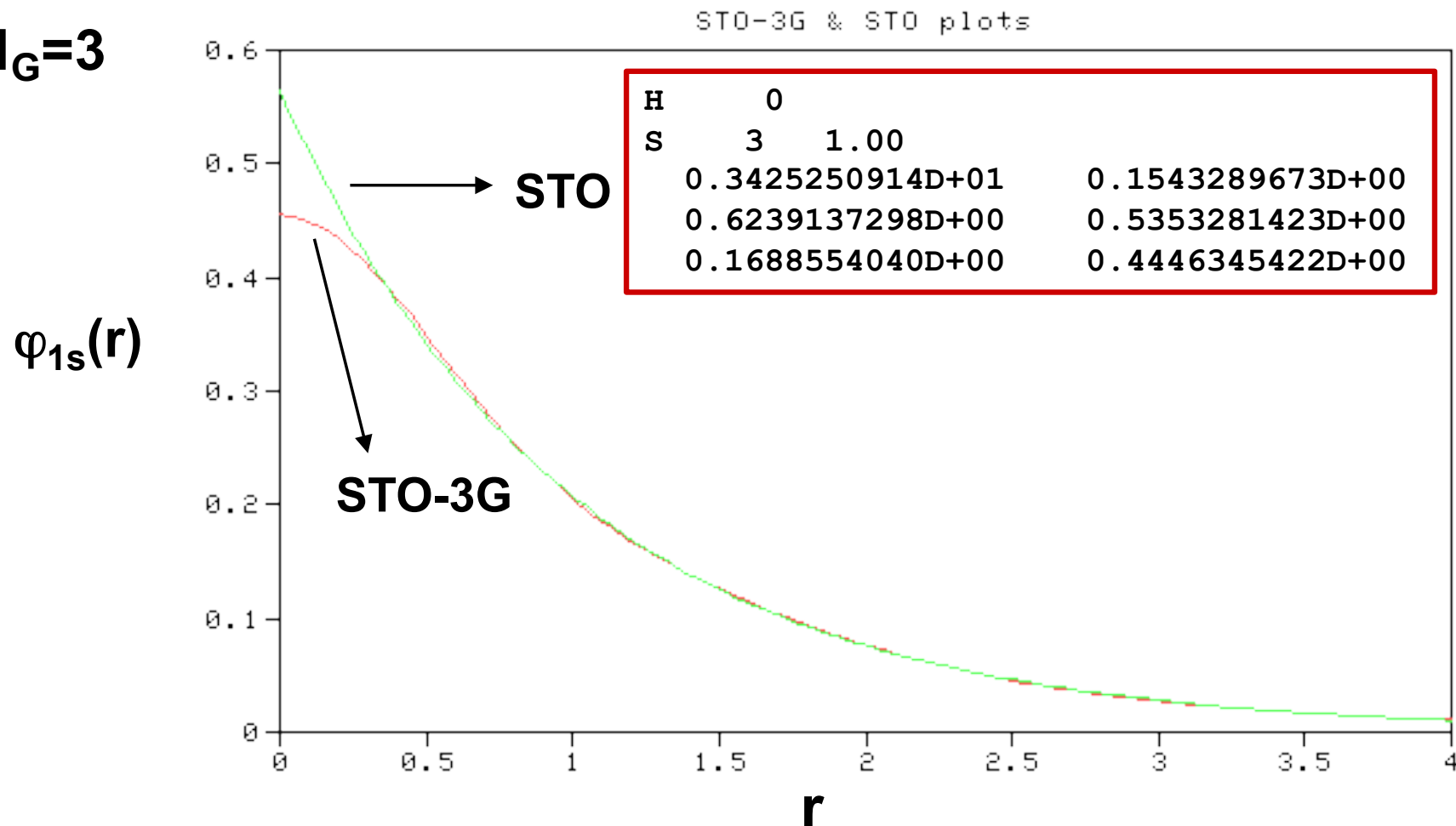
- This combination is called **Contracted Gaussian-type function**
- Primitive functions share the same center and angular momentum
- For each atomic species, the total basis is a finite set of contracted and uncontracted basis functions defined by the list of exponents and contraction coefficients of the primitive GTFs.

Example of Contracted Gaussian functions

Example: **H-atom 1s orbital**

$$\chi_{\mu}(\mathbf{r} - \mathbf{R}_{\mu}) = \sum_j^{N_G} d_{\mu,j} G(\alpha_{\mu,j}; \mathbf{r} - \mathbf{R}_{\mu})$$

$N_G=3$



Energy (exact): -0.5 Ha

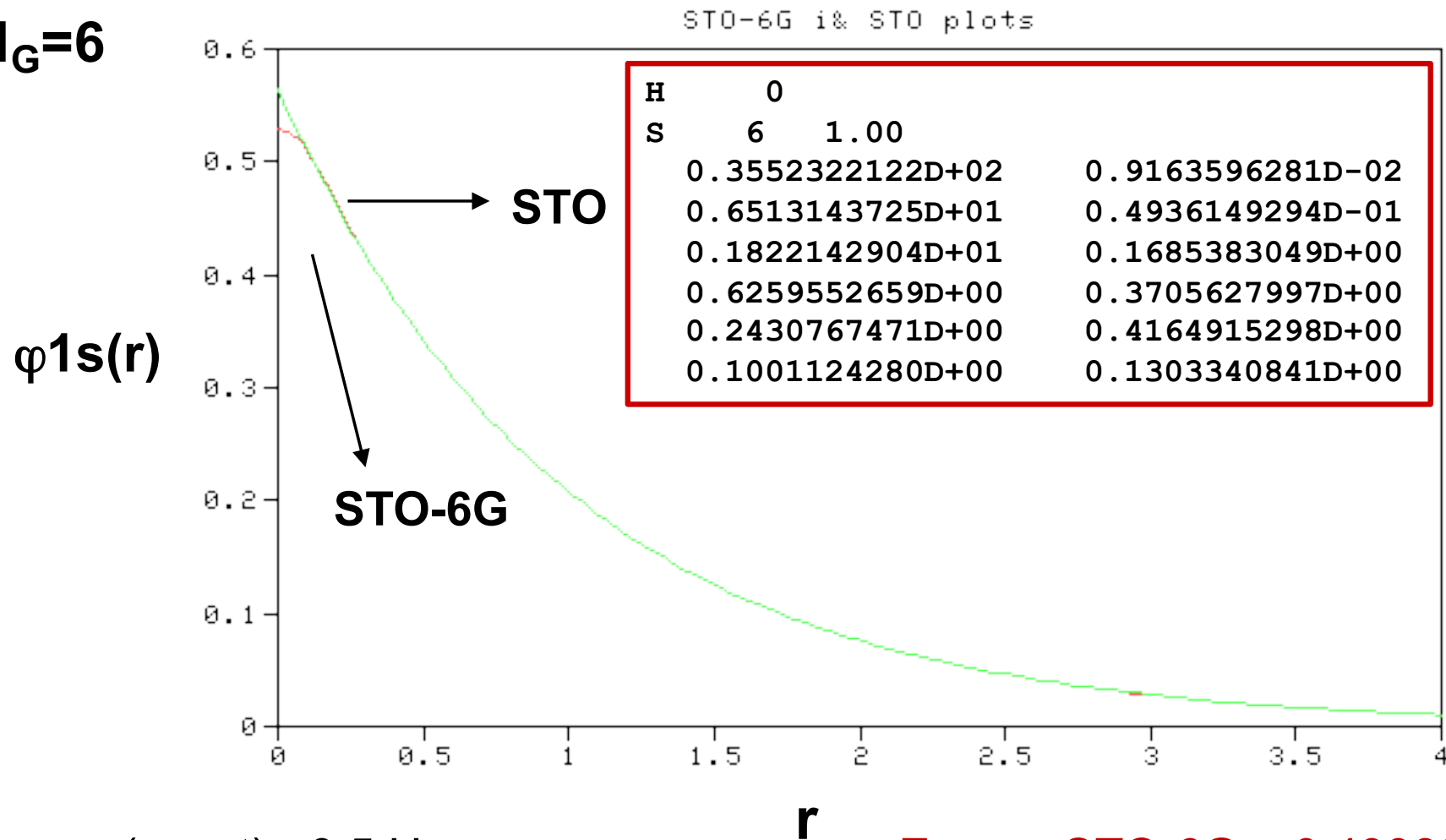
Energy **STO-3G**: -0.49491 Ha

Example of Contracted Gaussian functions

Example: **H-atom 1s orbital**

$$\chi_{\mu}(\mathbf{r} - \mathbf{R}_{\mu}) = \sum_j^{N_G} d_{\mu,j} G(\alpha_{\mu,j}; \mathbf{r} - \mathbf{R}_{\mu})$$

$N_G=6$



Energy (exact): -0.5 Ha

Energy **STO-6G**: -0.49983 Ha

Gaussian Basis Sets - **classification**

How many basis functions (N_{BS})?

- **Minimal basis set** (or single-zeta, **SZ**) basis has a single basis function for each orbital that is occupied in the atom

H: 1s

C: 1s, 2s, 2p_x, 2p_y, 2p_z

- The **Double-Zeta (DZ) basis set** consists of two basis functions with different exponents per atomic orbital and is twice as large as the minimal basis set.

H: 1s, 1s'

C: 1s, 1s', 2s, 2s', 2p_{x,y,z}, 2p_{x,y,z}'

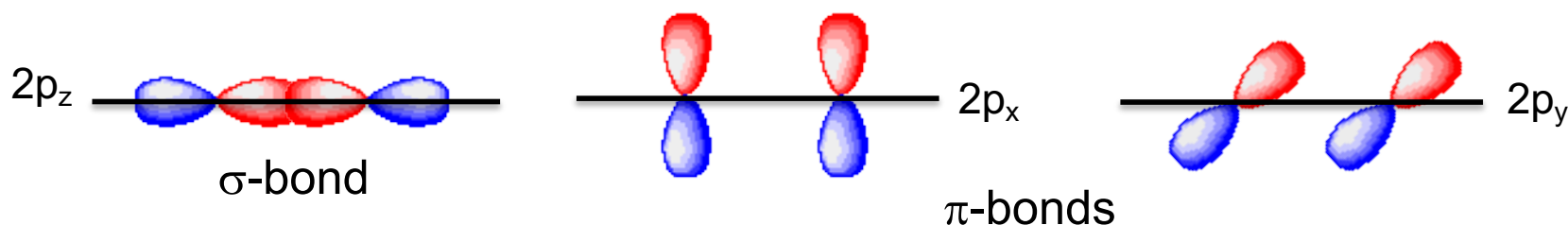
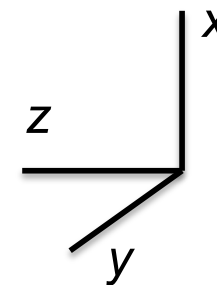
- This allows one to extend the basis set but it has also a crucial physical and chemical relevance
- Let's see an example...

Gaussian Basis Sets – Minimal vs Double Zeta

Example: C-C triple bond

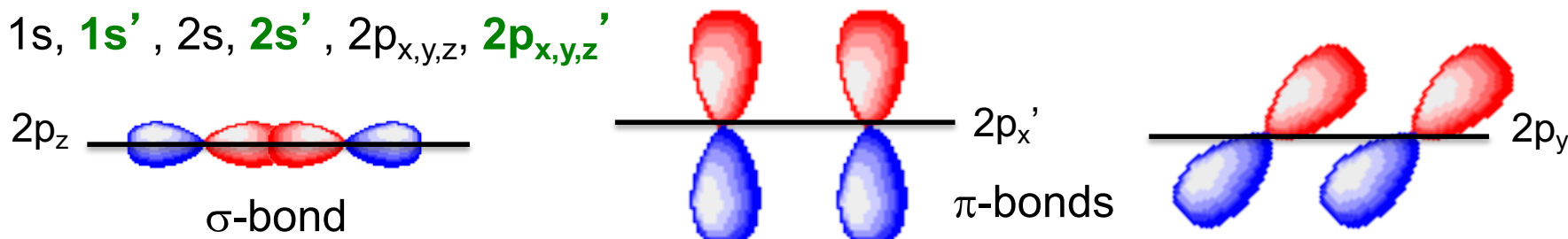
- Minimal basis set representation

C: 1s, 2s, **2p_x**, **2p_y**, **2p_z**



- The tighter $2p_z$ function is enough for the σ -bond, but more diffuse $2p_x$ and $2p_y$ functions would be better to describe the electron distribution of π -bonds
- Doubling all basis* functions gives more “flexibility” than the minimal BS.

C: 1s, **1s'**, 2s, **2s'**, $2p_{x,y,z}$, **2p_{x,y,z}'**



Gaussian Basis Sets - **classification**

How many basis functions (N_{BS})?

- **Minimal basis set** (or single-zeta, **SZ**) basis has a single basis function for each orbital that is occupied in the atom

H: 1s

C: 1s, 2s, 2p_x, 2p_y, 2p_z

- The **Double-Zeta (DZ) basis set** consists of two basis functions with different exponents per atomic orbital and is twice as large as the minimal basis set.

H: 1s, 1s'

C: 1s, 1s', 2s, 2s', 2p_{x,y,z}, 2p_{x,y,z}'

- The **Triple-Zeta (TZ) basis set** consists of three basis functions with different exponents per atomic orbital and is three times as large as the minimal basis set.

H: 1s, 1s', 1s''

C: 1s, 1s', 1s'', 2s, 2s', 2s'', 2p_{x,y,z}, 2p_{x,y,z}', 2p_{x,y,z}''

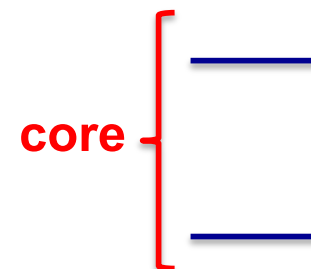
And so on: **QZ**, **5Z**, ...

Gaussian Basis Sets - classification

How many functions are needed, actually?



Idea: Chemical bonding occurs between valence electrons



- Doubling of the core orbitals is not needed, rarely adopted.
- Doubling is limited to valence orbitals
- The **Split-Valence** (SV) **basis set** is a minimal basis for core orbitals and is of double-zeta quality for the valence shells
- This leads to a **Valence** Double-Zeta (**VDZ**) or DZ split valence basis set (DZ**V**)
- The splitting can be applied to larger basis sets: TZ**V**, QZ**V**, ...

Gaussian Basis Sets – extensions (I)

Polarization functions are basis functions with angular momentum that is not occupied in the atom

Example:

- H, He: occupied s -orbitals $\Rightarrow$ p -type functions
- 1st-2nd rows: occupied p -orbitals $\Rightarrow$ d -type functions
- TM: occupied d -orbitals $\Rightarrow$ f -type functions
- Lanthanides: occupied f -orbitals $\Rightarrow$ g -type functions * New **CRYSTAL23**

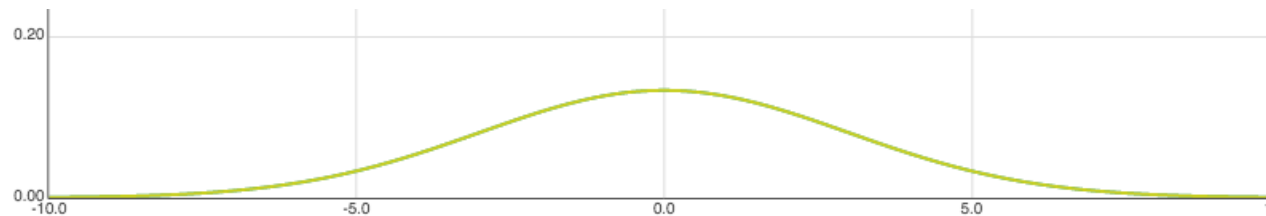


- They are important to describe polarized electronic charge distributions (ECD) (e.g. polar bonds) dipole moment, polarizability, dielectric response, anisotropy of ECD upon cell deformation, H-bonding, correlation energy (angular), ...
- The inclusion of a set of polarisation functions is often indicated by “**P**”
- Example: **DZVP**

* J.K. Desmarais, A. Erba, R. Dovesi, TCA (2018)

Gaussian Basis Sets – extensions (II)

Diffuse functions are usually *uncontracted* basis functions with small or very small exponents (e.g. $< 0.1 \text{ bohr}^{-2}$)



- They are useful to describe the tail of the electron charge distribution
- They can be important for anions, Rydberg states, metals, surfaces, weak interactions, ...
- The inclusion of a set of diffuse functions is often indicated by “+” or “D” or “aug-”

- **WARNING:**
 - For periodic systems very diffuse basis functions can cause numerical instabilities: $\Rightarrow$ avoid exponents lower than 0.07 ... 0.1 a.u.
 - Apart from specific cases, they are generally not necessary for solids



Gaussian Basis Sets – Notation

Typical notation:

(Primitive set) / [Contracted set]

(nr. of s-, p-, ... primitives) / [nr. of s-, p-, ... contractions]

E.g. (10s4p) / [3s2p]

In addition

{Contraction scheme}

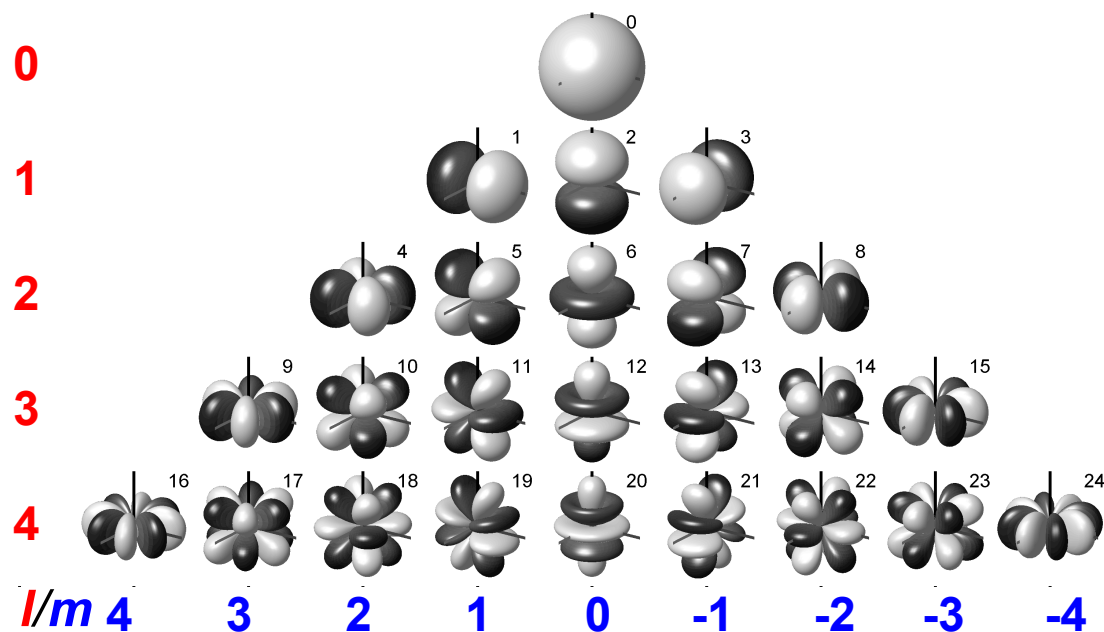
E.g. {631/31}

Usually, BSs are known by their names.

Examples of common families of molecular BSs:

Quality	Label	Name
SZ	Pople	STO-3G
DZ	Pople	3-21G
DZP	Jensen	pc-0
	Pople	6-31G*
	Ahlrichs	Def2-SV(P)
	Ahlrichs	Def2-SVP
	Jorge	DZP
	Sapporo	DZP
	Roos	ANO-DZP
	Dunning	cc-pVDZ ¹
	Petersson	2ZaP
	Jensen	pc-1
TZP	Pople	6-311G(2df)
	Ahlrichs	Def2-TZVP
	Ahlrichs	Def2-TZVPP
	Jorge	TZP
	Sapporo	TZP
	Roos	ANO-TZP
	Dunning	cc-pVTZ
	Petersson	3ZaP
	Jensen	pc-2
	Ahlrichs	Def2-QZVP
QZP	Ahlrichs	Def2-QZVPP
	Sapporo	QZP
	Roos	ANO-QZP
	Dunning	cc-pVQZ
	Jensen	pc-3
	Dunning	cc-pV5Z
5ZP	Jensen	pc-4

Angular part and shells



g functions are now available in **CRYSTAL23**

Atomic basis sets are specified according to the number and type of shells

How the orbitals are ordered is important for electronic state occupation
(keyword: **EIGHSHIFT, FDOCCUP**)
To compute properties as population analysis, density of states (keyword: **PPAN, DOSS**)

shell code	shell type
0	S
1	SP
2	P
3	D
4	F
5	G

n.
AO

order of internal storage

s
 s, x, y, z
 x, y, z
 $2z^2 - x^2 - y^2, xz, yz, x^2 - y^2, xy$
 $(2z^2 - 3x^2 - 3y^2)z, (4z^2 - x^2 - y^2)x, (4z^2 - x^2 - y^2)y,$
 $(x^2 - y^2)z, xyz, (x^2 - 3y^2)x, (3x^2 - y^2)y$
 $3x^4 + 6x^2y^2 - 24x^2z^2 + 3y^4 - 24y^2z^2 + 8z^4, (4z^2 - 3x^2 - 3y^2)xz,$
 $(4z^2 - 3x^2 - 3y^2)yz, 6x^2z^2 - x^4 + y^4 - 6y^2z^2, (6z^2 - x^2 - y^2)xy$
 $(x^2 - 3y^2)xz, (3x^2 - y^2)yz, x^4 - 6x^2y^2 + y^4, (x^2 - y^2)xy$

Basis set input section - I

E.g.: MgO bulk - fcc cubic cell – RHF/STO-3G

Mg

```
12 3
0 0 3 2. 1.
2.992E+02 1.543E-01
5.451E+01 5.353E-01
1.475E+01 4.446E-01
0 1 3 8. 1.
1.512E+01-9.997E-02 1.559E-01
3.514E+00 3.995E-01 6.077E-01
1.143E+00 7.001E-01 3.920E-01
0 1 3 2. 1.
1.395E+00-2.196E-01 1.059E-02
3.893E-01 2.256E-01 5.952E-01
1.524E-01 9.004E-01 4.620E-01
8 2
0 0 3 2. 1.
1.307E+02 1.543E-01
2.381E+01 5.353E-01
6.444E+00 4.446E-01
0 1 3 6. 1.
5.033E+00-9.997E-02 1.559E-01
1.170E+00 3.995E-01 6.077E-01
3.804E-01 7.001E-01 3.920E-01
99 0
[Optional keywords]
END
```

Atomic number and number of shells

End of basis set input section

Basis set input section - I

E.g.: MgO bulk - fcc cubic cell – RHF/STO-3G

Mg

12 3

0 0 3 2. 1.

2.992E+02 1.543E-01

5.451E+01 5.353E-01

1.475E+01 4.446E-01

0 1 3 8. 1.

1.512E+01-9.997E-02 1.559E-01

3.514E+00 3.995E-01 6.077E-01

1.143E+00 7.001E-01 3.920E-01

0 1 3 2. 1.

1.395E+00-2.196E-01 1.059E-02

3.893E-01 2.256E-01 5.952E-01

1.524E-01 9.004E-01 4.620E-01

8 2

0 0 3 2. 1.

1.307E+02 1.543E-01

2.381E+01 5.353E-01

6.444E+00 4.446E-01

0 1 3 6. 1.

5.033E+00-9.997E-02 1.559E-01

1.170E+00 3.995E-01 6.077E-01

3.804E-01 7.001E-01 3.920E-01

99 0

[Optional keywords]

END

Atomic number and number of shells
Type of shell, nr. primitives, occupation
Exponents and contraction coefficients

End of basis set input section

Basis set errors

Basis Set Balancing Error (BSBE)

- Atomic basis sets must be of the same quality both in the same system and when comparing energies of two systems (trivial)

Basis Set Incompleteness Error (BSIE)

- A near-complete basis set is unaffordable – too many basis functions
- Complete Basis Set (CBS) limit => hierarchical BS & extrapolation
- No problem for molecule - more delicate for solids

Basis Set Superposition error (BSSE)

- Artificial overestimation of binding energies (interatomic, inter- & intra-molecular)
- More relevant for densely packed solids and small basis sets
- Corrections are needed (Counterpoise (CP) method, geometrical CP)



Basis set superposition error (BSSE)

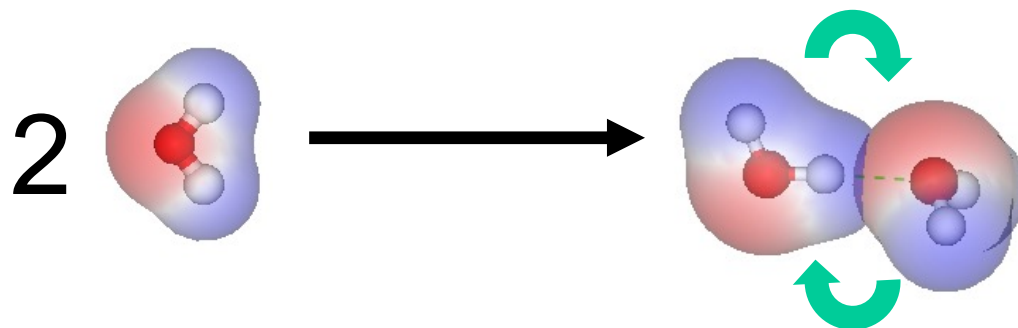
In calculation on system AB with a *finite* basis set, the description of fragment A will be improved by the functions on fragment B, and viceversa

This effect will lower the energy, implying extra binding between A and B. It may even suggest binding where there is none

Most commonly estimated and analyzed (a posteriori) via the *counterpoise correction*: compute the fragment energies not in the individual basis set (A or B) but in the total basis set (A[B], B[A] – *ghost functions*)

BSSE: GTOs vs PWs energies

GTOs



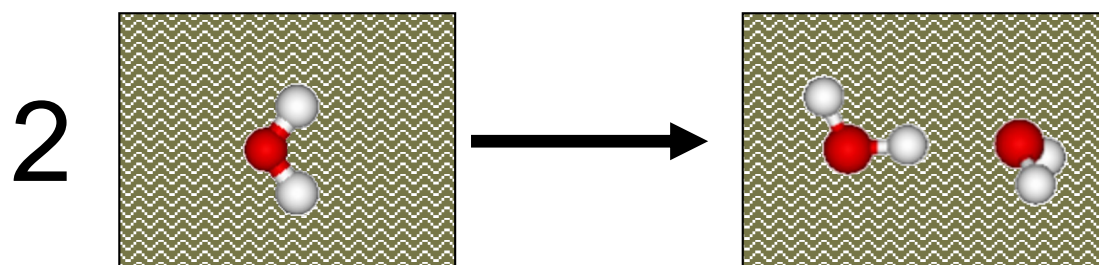
The energy of the dimer is artificially lowered by exploiting the partner basis set

$$\Delta E = E(\text{dimer}) - 2 \epsilon(\text{monomer})$$

$$\text{BSSE} = \Delta E - \Delta E^C$$

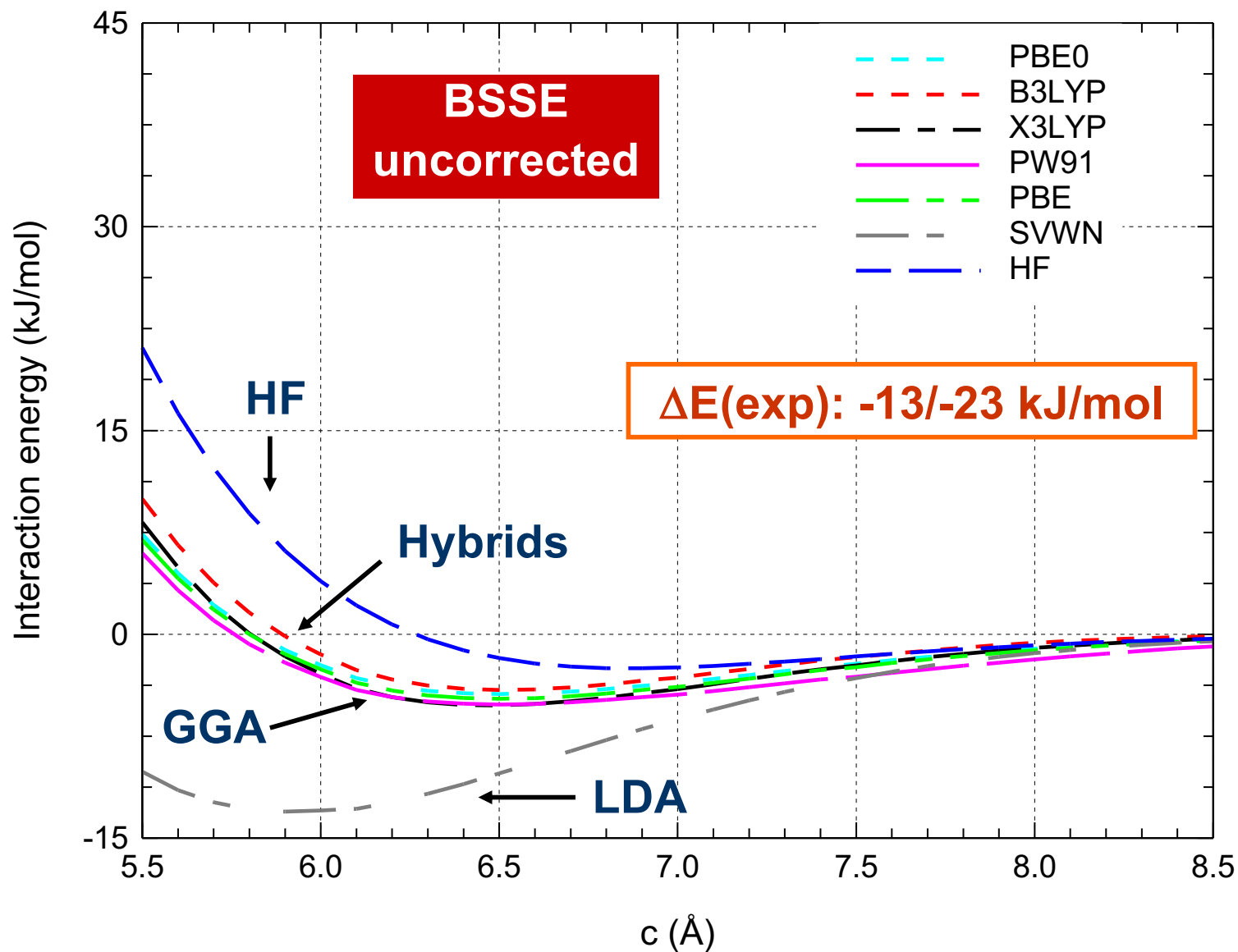
$$\Delta E^C = E(\text{dimer}) - \epsilon_1(\text{monomer}) - \epsilon_2(\text{monomer})$$

PWs



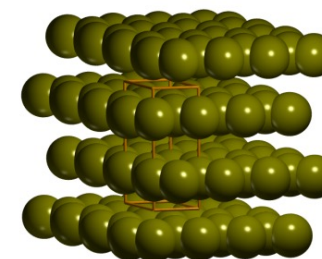
PWs fill the whole box so energies are BSSE free

Interlayer interaction in graphite

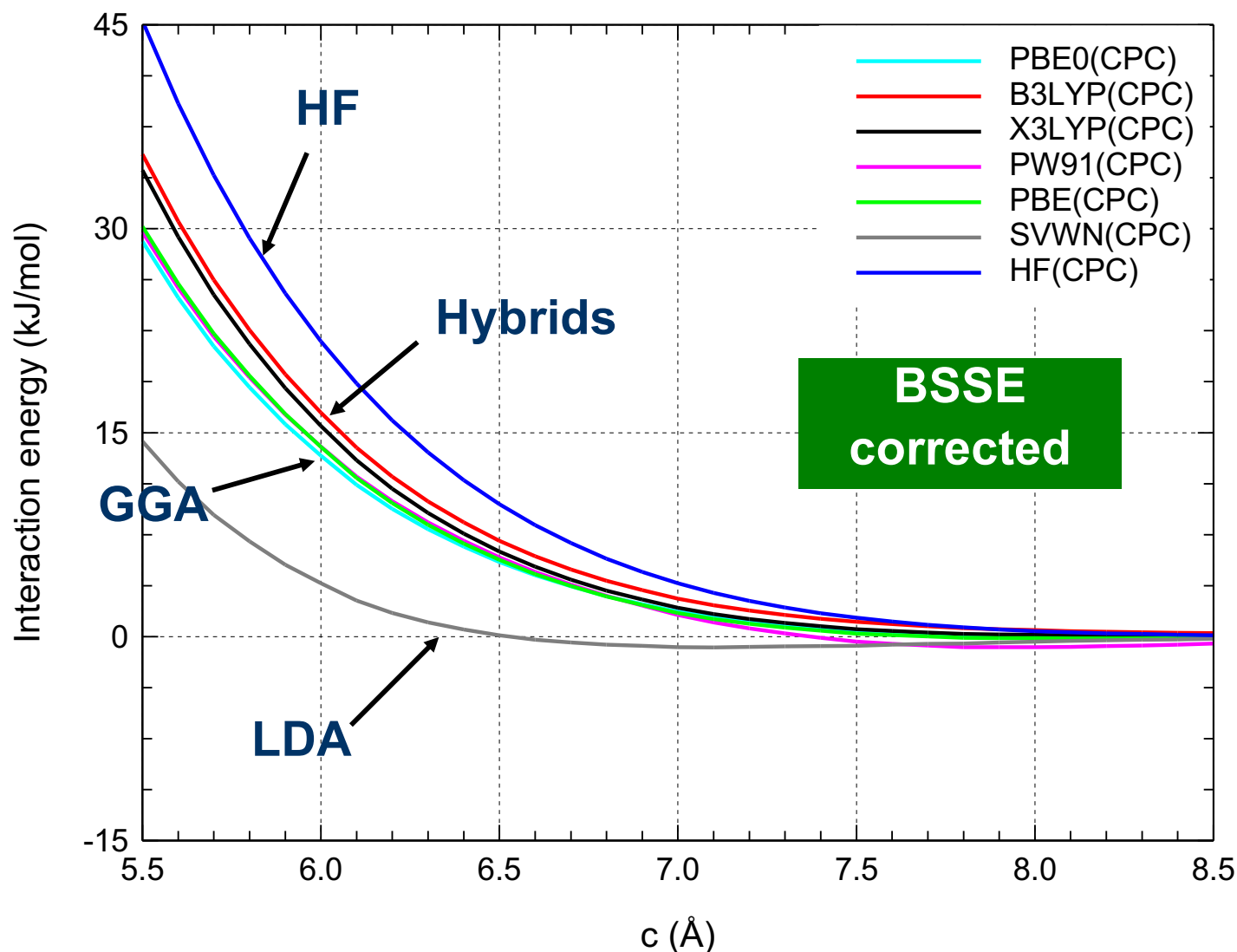


Exp.:
 $a = 2.46$ (fixed)
 $c = 6.71 \text{ Å}$

BS: 6-31G(d)

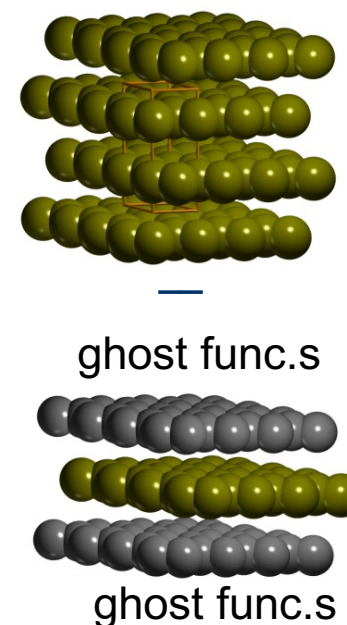


Interlayer interaction in graphite



Exp.:
 $a = 2.46$ (fixed)
 $c = 6.71$ Å

BS: 6-31G(d)



BSSE can give binding where there is none.
HF and DFT functionals do not account for dispersion energy

How to choose a basis set

**Balance between accuracy and computational cost.
But, computational cost should not determine what basis set is used.**



Molecular basis sets form a good starting point:

- well defined,
- databases,
- hierarchy (e.g. cc-pVXZ, $X=D,T,Q,5,6$)).

However, unmodified molecular basis sets are not fully suitable for solid state calculations

For periodic systems, *basis set optimization* is usually necessary:

- Larger variety of binding, basis sets are less transferable
- *All-purpose vs system-specific* optimized basis sets

Libraries are available

Gaussian basis set databases

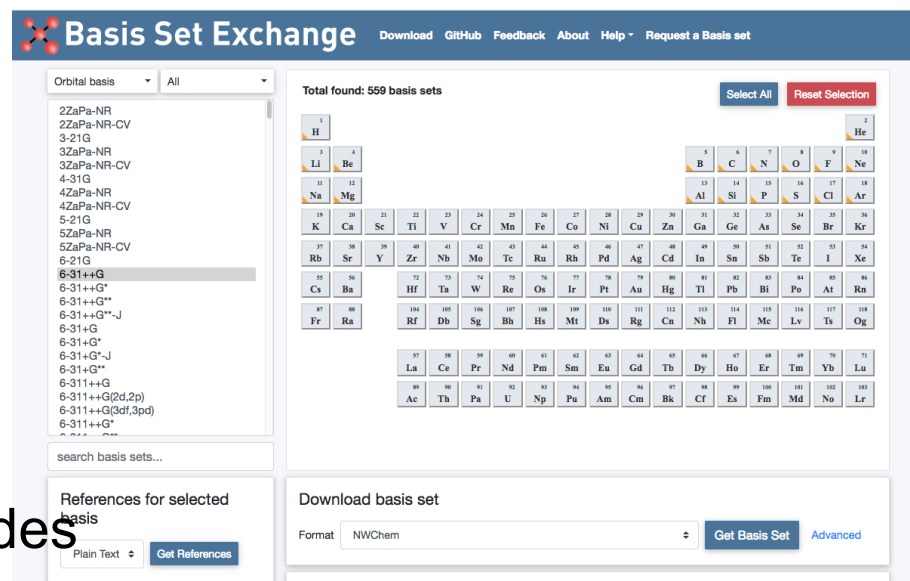
Molecular basis sets:

Basis set exchange database

<https://www.basissetexchange.org/>

A huge number of basis sets

See also the web site of molecular codes



The screenshot shows the Basis Set Exchange website. On the left, there is a list of basis sets under the 'Orbital basis' dropdown, which is set to 'All'. The list includes various sets like 2ZaPa-NR, 3-21G, 4-31G, 5-21G, 6-31+G, etc. Below the list is a search bar and a 'References for selected basis' section. On the right, a periodic table is displayed with elements colored by their basis set. Above the table, it says 'Total found: 559 basis sets'. Below the table, there is a 'Download basis set' section with a 'Format' dropdown set to 'NWChem' and a 'Get Basis Set' button.

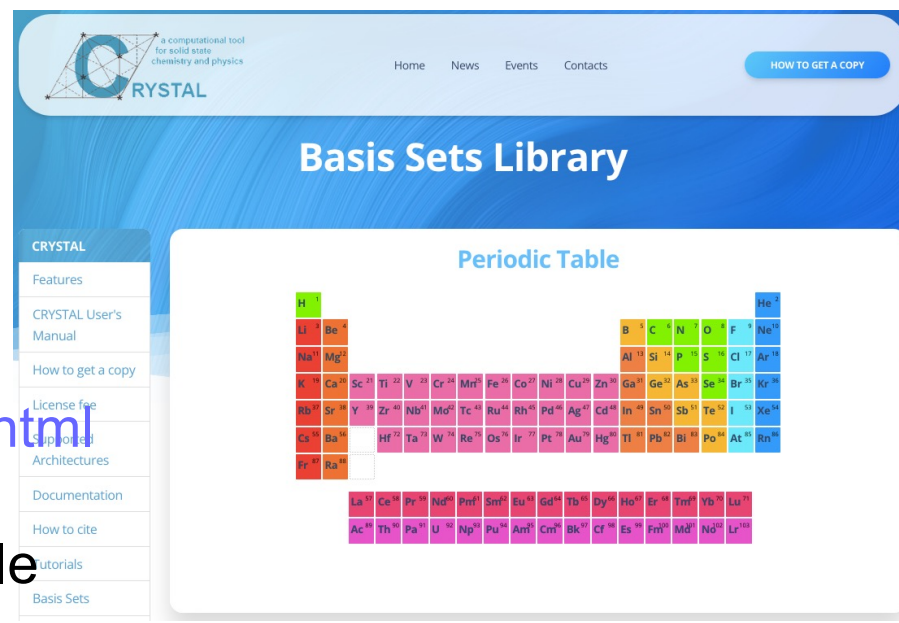
Crystalline solids:

CRYSTAL web site

A basis set library is available at:

https://www.crystal.unito.it/basis_sets.html

Almost all elements of the periodic table



The screenshot shows the CRYSTAL Basis Sets Library website. The header includes the CRYSTAL logo and navigation links: Home, News, Events, Contacts. A 'HOW TO GET A COPY' button is also present. The main section is titled 'Basis Sets Library' and features a 'Periodic Table' with elements colored by their basis set. On the left, there is a sidebar with links to 'Features', 'CRYSTAL User's Manual', 'How to get a copy', 'License fee', 'Supported Architectures', 'Documentation', 'How to cite', 'Tutorials', and 'Basis Sets'.

New basis sets and tools in CRYSTAL23

All-purpose POB DZVP
and TZVP basis sets have
been extended and
revised (rev2 – less BSSE)
Elements from H to Po
(Bredow & co. JCC 2013, 2019, 2021)

Basis sets (*f in valence*)
for *Lanthanides and Actinides*
have been developed.

(El Kalani, et al. PRB (2018))

An automated algorithm (BDIIS) for *system-specific* basis set optimization has been implemented.

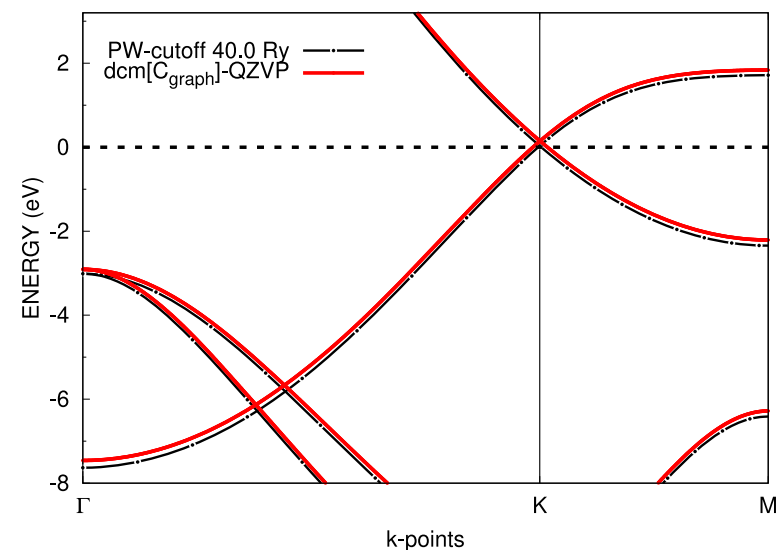
Keyword: **BSOPT** (*advanced option*)

(e.g. Graphene band structure PW vs QZVP)

(L. Daga, BC, L. Maschio JCTC 2020)

1 H Hydrogen 1											2 He Helium 4						
3 Li Lithium 3	4 Be Beryllium 9											5 B Boron 11	6 C Carbon 12	7 N Nitrogen 14	8 O Oxygen 16	9 F Fluorine 19	10 Ne Neon 20
11 Na Sodium 23	12 Mg Magnesium 24											13 Al Aluminum 27	14 Si Silicon 28	15 P Phosphorus 31	16 S Sulfur 32	17 Cl Chlorine 35.5	18 Ar Argon 40
19 K Potassium 39	20 Ca Calcium 40	21 Sc Scandium 45	22 Ti Titanium 48	23 V Vanadium 51	24 Cr Chromium 52	25 Mn Manganese 55	26 Fe Iron 56	27 Co Cobalt 59	28 Ni Nickel 59	29 Cu Copper 64	30 Zn Zinc 65	31 Ga Gallium 70	32 Ge Germanium 73	33 As Arsenic 75	34 Se Selenium 79	35 Br Bromine 80	36 Kr Krypton 84
37 Rb Rubidium 86	38 Sr Strontium 88	39 Y Yttrium 89	40 Zr Zirconium 91	41 Nb Niobium 93	42 Nb Niobium 96	43 Tc Technetium 98	44 Ru Ruthenium 101	45 Rh Rhodium 103	46 Pd Palladium 106	47 Ag Silver 108	48 Cd Cadmium 112	49 In Indium 115	50 Sn Tin 119	51 Sb Antimony 122	52 Te Tellurium 128	53 I Iodine 127	54 Xe Xenon 131
55 Cs Cesium 133	56 Ba Barium 137	57 La Lanthanum 139	72 Hf Hafnium 179	73 Ta Tantalum 181	74 W Tungsten 184	75 Re Rhenium 186	76 Os Osmium 190	77 Ir Iridium 192	78 Pt Platinum 195	79 Au Gold 197	80 Hg Mercury 201	81 Tl Thallium 204	82 Pb Lead 207	83 Bi Bismuth 209	84 Po Polonium 210	85 At Astatine 210	86 Rn Radon 222
87 Fr Francium 223	88 Ra Radium 226	89 Ac Actinium 227	104 Unq Unquadrium 257	105 Unp Unpentium 260	106 Unh Unhexium 263	107 Uus Unseptium 266	108 Uno Unoctium 269	109 Uue Unennium 272									

58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
Cerium 140	Praseodymium 141	Neodymium 144	Promethium 147	Samarium 150	Europium 152	Gadolinium 157	Terbium 159	Dysprosium 163	Holmium 165	Erbium 167	Thulium 169	Ytterbium 173	Lutetium 175
90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr
Thorium 232	Protactinium 231	Uranium 238	Neptunium 237	Plutonium 244	Americium 243	Curium 247	Berkelium 247	Californium 249	Einsteinium 254	Fermium 253	Hassium 256	Nobelium 254	Lawrencium 257



Basis set input section - II

E.g.: MgO bulk - fcc cubic cell – RHF/STO-3G

Mg

```
12 3
0 0 3 2. 1.
2.992E+02 1.543E-01
5.451E+01 5.353E-01
1.475E+01 4.446E-01
0 1 3 8. 1.
1.512E+01-9.997E-02 1.559E-01
3.514E+00 3.995E-01 6.077E-01
1.143E+00 7.001E-01 3.920E-01
0 1 3 2. 1.
1.395E+00-2.196E-01 1.059E-02
3.893E-01 2.256E-01 5.952E-01
1.524E-01 9.004E-01 4.620E-01
8 2
0 0 3 2. 1.
1.307E+02 1.543E-01
2.381E+01 5.353E-01
6.444E+00 4.446E-01
0 1 3 6. 1.
5.033E+00-9.997E-02 1.559E-01
1.170E+00 3.995E-01 6.077E-01
3.804E-01 7.001E-01 3.920E-01
99 0
[Optional keywords]
END
```

O

Internal BS library

MgO Fm-3m ICSD 240598

0 0 0

225

4.21

2

12 0.0 0.0 0.0

8 0.5 0.5 0.5

BASISSET

STO-3G

SHRINK

8 8

END

Effective Core Potentials (ECP)

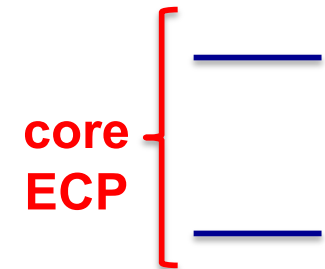
Do we really need to explicitly treat core electrons for heavy metals?

Idea: reduce computational effort with an **effective potential** which describes nucleus and inner electrons + include a part of the **relativistic effects**



Shape-consistent: fit shape of valence orbitals to all-electron shape

Energy-consistent: fit to a set of energies of electronic states of an atom/ion



Relativistic effects can be included in the ECP at different levels of approximations from simplified (e.g. Wood-Boring) to fully relativistic (4-component Dirac-Fock)

Effective one-electron operator:

$$\hat{h}(i) = -\frac{1}{2}\Delta_i + V_{PP}(i)$$

Effective core potential:

$$V_{PP}(i) = -\frac{Q}{r} + \sum_{l,j} [\sum_k B_{lj,k} e^{-\beta_{lj,k} r^2}] \mathcal{P}_{lj}$$

Library of ECPs

Stuttgart/Cologne energy-consistent (ab initio) pseudopotentials

<http://www.tc.uni-koeln.de/PP/index.en.html>

Available in CRYSTAL format

Labeled as: ECP_nXY

n = nr. of core electrons

XY = type of fit and relativistic

e.g. ECP10MWB :

nucleus + $1s^2 2s^2 2p^6$

replaced by

pseudopotential (“Ne core”,
10 electrons)

Multi-valence-electron atom
used in fit (M),
relativistic (WB)

Energy-consistent Pseudopotentials of the Stuttgart/Cologne Group

Chose the data format and click an element in the periodic table to get the pseudopotential parameters and corre

Data format:

Crystal09

Periodic system of elements:

Click on the element for which you want a pseudopotential and a basis set !

H																				He
Li	Be											B	C	N	O	F	Ne			
Na	Mg											Al	Si	P	S	Cl	Ar			
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr			
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe			
Cs	Ba	Ln	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn			
Fr	Ra	An	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Uu	Fl	Up	Lv	Us	Uo			
Un	Ux																			
Ln=	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu					
An=	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr					

Example of ECP: ECP28MWB for Sr

Pseudopotential ECP28MWB for Sr

ECP for Sr

COMMENT # Q=10., MEFIT, WB, Ref 13.

COMMENT # Sr

238 <here, the number of valence-basis shells is needed>

INPUT

10. 0 2 2 2 1 0

7.400074 135.479430 0

3.606379 17.534463 0

6.484868 88.359709 0

3.288053 15.394372 0

4.622841 29.888987 0

2.246904 6.659414 0

4.633975 -15.805992 0

<please add the valence-basis input, without blank line>

COMMENT # References:

COMMENT # [13] M. Kaupp, et. Al., J. Chem. Phys. 94, 1360 (1991).

Available valence basis sets for Sr, ECP28MWB:

ECP28MWB

ECP28MDF_GUESS (ECP28MHF_GUESS ECP28MWB_GUESS)

Core: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$

28 electrons in core

10 valence electrons

Note: Use ECP data as is – no change when used with CRYSTAL

Hints for classes of crystalline systems

Covalent systems: molecular basis sets can be sometimes used, reoptimisation of outer exponents is suggested

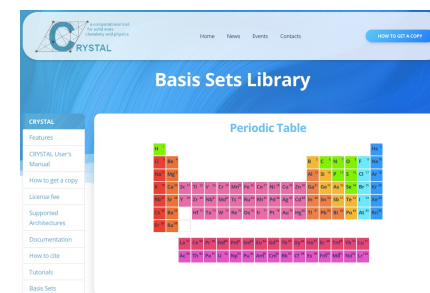
Ionic / Semi-ionic systems: reoptimisation of the most diffuse valence shells is mandatory, in particular of anions

Metals: diffuse functions are necessary, reoptimisation is recommended

Molecular crystals: molecular basis sets can be used largely unmodified, care needed to too diffuse functions

Surfaces and Adsorption: diffuse functions could be needed, molecular basis sets can be used for the adsorbate

Refer to libraries of basis set for crystalline systems:
<http://www.crystal.unito.it> $\Rightarrow$ Basis sets



Gaussian BS for solids – final remarks

- The ultimate accuracy of any calculation is determined by the adopted basis set
- Libraries of bespoke basis sets are available as starting point
- New (or molecular) basis sets may need to be optimised
- Basis set must have sufficient flexibility
 - Need to consider:
 - Number and type of functions per shell
 - Number of shells per atom/ion
- Care needed to achieve convergence of quantity of interest.
 - Some properties are easier than others.
 - Try to use a hierarchy of BS (e.g. DZ => TZ)

Suggested readings

R. Dovesi, B. Civalleri, R. Orlando, C. Roetti, and V.R. Saunders **“*Ab Initio Quantum Simulation in Solid State Chemistry*”** *Rev. Comp. Chem.* 21 (2005) 1

Klaus Doll **“Gaussian Basis Sets for Solid State Calculations”** pag. 157-181, in *Lecture Notes in Chemistry*, vol. 107 (2021)

Mike Towler **“An introductory guide to Gaussian basis sets in solid-state electronic structure calculations”** *Notes for MSSC2000*

F. Jensen **“Atomic orbital basis sets”** *WIREs Comput Mol Sci* 2013, 3: 273–295 doi: 10.1002/wcms.1123

M. Dolg and X. Cao **“Relativistic Pseudopotentials: Their Development and Scope of Applications”** *Chem. Rev.* 112 (2012) 403

Acknowledgements

Part of the material presented has been adapted from slides kindly provided by K. Doll, N.M. Harrison, M. Peintinger, T. Laun, ...

